Python for Supply Chain

Session 3 · Mon 23 Nov 2026

Let the loop do the walking

Today's question

Can one rule label every order line?

A for loop walks down the list

IN DAILY LIFE

Walking down the shelf, checking each bin.

IN EXCEL

Dragging a formula down.

THE LIMIT

The body must be indented.

```
quantities = [6, 6, 8, 6, 6, 2, 6]
for quantity in quantities:
    print(quantity)
print("Done")
```

A function is your own reusable rule

IN DAILY LIFE

Your own reusable rule.

IN EXCEL

A named formula.

THE LIMIT

It does nothing until you call it.

```
def order_size(quantity):
    if quantity >= 100:
        return "big"
    elif quantity >= 12:
        return "medium"
    return "small"
```

print shows, return hands the answer back

IN DAILY LIFE

Showing the answer, or handing it over.

IN EXCEL

The value a formula shows.

THE LIMIT

print only shows. return hands it back.

```
def order_size_printed(quantity):
    print("big")

print(order_size_printed(1930))
print(order_size(1930))
```

Read a traceback from the bottom

Read	What you get
Last line	The error type and message
Line above	The line that broke
Higher up	Who called that line

Start at the last line. Then walk up.

Now in Colab

Notebook	s3_let_loop_walk.ipynb
In class	Loop, function, return, traceback
Real order	Invoice 540815: 1,930, 2,000, 3,114 units

Three things you can do now

Walk a list

for quantity in quantities

Wrap a rule

def order_size(quantity)

Hand back an answer

return, not print

Homework and daily recall

Homework

1. Write line_value(quantity, price)
2. Loop over the 5 handout lines
3. Write one function of your own
4. Mount Drive once at home

Daily recall

1. Three questions Tuesday 24
2. Three questions Wednesday 25